105120

AMYLOIDOSIS, FINNISH TYPE

Alternative titles; symbols

AMYLOIDOSIS, HEREDITARY SYSTEMIC 4, FINNISH TYPE; AMYLD4
AMYLOIDOSIS V
AMYLOIDOSIS, MERETOJA TYPE
AMYLOID CRANIAL NEUROPATHY WITH LATTICE CORNEAL DYSTROPHY
AMYLOIDOSIS DUE TO MUTANT GELSOLIN

ORPHA: 85448; **DO:** 0050637; **MONDO:** 0007097;

Phenotype-Gene Relationships

Location	Phenotype	Phenotype MIM number	Inheritance	Phenotype mapping key	Gene/Locus	Gene/Locus MIM number
9q33.2	Amyloidosis, Finnish type	105120	Autosomal dominant	3	GSN	137350

TEXT

A number sign (#) is used with this entry because of evidence that the Finnish type of amyloidosis is caused by heterozygous or homozygous mutation in the gelsolin gene (GSN; 137350) on chromosome 9q33.

Description

The Finnish type of systemic amyloidosis is characterized clinically by a unique constellation of features including lattice corneal dystrophy, and cranial neuropathy, bulbar signs, and skin changes. Some patients may develop peripheral neuropathy and renal failure. The disorder is usually inherited in an autosomal dominant pattern; however, homozygotes with a more severe phenotype have also been reported (Meretoja, 1973).

Finnish hereditary amyloidosis, also known as Meretoja syndrome or AGel amyloidosis, is one of the most common diseases in the Finnish disease heritage. Symptoms commonly appear by age 40, with the first finding usually corneal lattice dystrophy (CLD), diagnosed by an ophthalmologist. Impaired vision, polyneuropathy, facial nerve paresis, and cutis laxa follow. These symptoms may develop slowly and simultaneously, since amyloid accumulates systemically at a constant rate (summary by Nikoskinen et al., 2015).

For a discussion of genetic heterogeneity of hereditary systemic amyloidosis, see AMYLD1 (105210).

Clinical Features

In a massive investigation in Finland, Meretoja (1973) identified 207 affected persons. Two patients, whose parents were affected and who were more severely affected than the others, were thought to represent homozygosity. A few of the patients developed nephrotic syndrome and renal failure and some had cardiac involvement. Amyloid involvement was rather widespread at autopsy. Meretoja et al. (1978) collected 307 patients in Finland.

Klintworth (1967) presented evidence that corneal dystrophy of the lattice type is a local variety of amyloidosis. Lattice corneal dystrophy accompanied systemic amyloidosis of the Finnish type.

Three Czechoslovakian sisters with bulbar palsy, 'cutis hyperelastica,' and lattice dystrophy of the cornea, reported by Klaus et al. (1959), may have had this disorder. Cases were reported from the United States by Sack et al. (1981), Purcell et al. (1983), Darras et al. (1986), and Starck et al. (1991); from Holland by Winkelman et al. (1971); and from Denmark by Boysen et al. (1979).

One of the patients reported by Sack et al. (1981) had onset of facial paralysis, which began as inability to control a drooping lower lip, at the age of about 56; the lip became strikingly protuberant and everted with exposure of the lower gingival mucosa. Five years after onset he could not wrinkle his forehead and there was an intermittent twitch of the right side of the lower lip. The extraocular muscles were affected only minimally and there was no ptosis. A striking feature was laxity of the skin, which raised the question of cutis laxa. Slit-lamp examination showed a lattice type of corneal opacity bilaterally. The mother had the identical disorder beginning at about the same stage of life. The proband had bulbar manifestations.

Kiuru (1992) reported the clinical findings of 30 patients. Signs of cranial neuropathy especially affecting the facial nerve were found in all, and peripheral polyneuropathy mainly affecting vibration and touch senses was demonstrated in 26 patients. Kiuru et al. (1994) studied the autonomic nervous system and heart in 30 patients. Minor autonomic nervous system dysfunction was found in most patients, but clinically

significant autonomic dysfunction or cardiopathy was not characteristic.

Akiya et al. (1996) reported a Japanese brother and half sister with lattice corneal dystrophy as part of Finnish-type amyloidosis. The authors referred to the Finnish type as FAP type IV. The patients were 70 and 68 years old.

Potrc et al. (2021) reported 6 members of a family with Finnish-type amyloidosis. Family members had corneal dystrophy (5/6), loose skin (5/6), and cardiac arrhythmia requiring treatment with an implantable cardioverter defibrillator (3/6). One patient had bilateral optic neuropathy, which may be related to the disease and is critical to recognize before corneal transplantation. Histopathologic analysis of the eyelid skin showed atrophic epidermis and dermis and amyloid deposits around hair follicles.

Mullany et al. (2021) reported 2 unrelated families with Finnish-type amyloidosis, one with an affected female without a family history of the disease and the other with affected father, son, and daughter. All 4 patients had polymorphic corneal stromal opacities, dermatochalasis requiring upper eyelid blepharoplasty, and dry eyes. The father and son had open-angle glaucoma. Systemic features included cutis laxa in all 4 patients, proteinuria in the daughter, cardiac findings in the father (sinus bradycardia) and the daughter (supraventricular tachycardia and idiopathic cardiomyopathy), and hypothyroidism in the son and daughter.

Inheritance

The transmission pattern of the Finnish type of amyloidosis in the family reported by Maury et al. (1990) was consistent with autosomal dominant inheritance.

Nomenclature

The type of corneal lattice dystrophy (CLD) that occurs as a feature of Finnish amyloidosis has been referred to as corneal lattice dystrophy type II (CLD2) or lattice corneal dystrophy type II (LCD2).

Akiya et al. (1996) stated that lattice corneal dystrophy type II (LCD2) is a characteristic finding of Finnish amyloidosis.

Nikoskinen et al. (2015) noted that corneal lattice dystrophy (CLD2) is usually the first symptom of Finnish amyloidosis to appear, is 'practically prerequisite,' and with positive family history is sufficient for diagnosis of Finnish amyloidosis.

Schmidt et al. (2020) stated that corneal lattice amyloidosis had been incorrectly referred to as lattice corneal dystrophy type II and is the first ophthalmologic sign to appear.

Molecular Genetics

Maury et al. (1990) studied amyloid fibrils isolated from the kidney of a patient with the Finnish form. The amino acid sequence determined for part of the protein was identical to that deduced for plasma gelsolin in the region of amino acids 235-269. Using PCR and allele-specific oligonucleotide hybridization analysis of genomic DNA in patients with this disorder, Maury et al. (1990) identified a 654G-A transition in the GSN gene, resulting in an asp187-to-asn substitution (D187N; 137350.0001), in all 5 unrelated patients studied, but not in 45 unrelated control subjects. In a family study, the mutant gene was demonstrated in all members with clinical symptoms of the disease, as well as in 1 asymptomatic family member who was thought to represent a presymptomatic gene carrier.

Haltia et al. (1990) likewise showed that the amyloid in this disorder is antigenically and structurally related to gelsolin. The same mutation in gelsolin (asp187-to-asn) has been found in all Finnish families studied to date (Maury et al., 1990; Paunio et al., 1992; de la Chapelle et al., 1992; Haltia et al., 1992; Sipila and Aula, 2002); furthermore, it was also found in the affected son of the proband of the Scottish-American family reported by Sack et al. (1981); see de la Chapelle et al. (1992).

Maury (1993) reported the findings in 2 sisters who were homozygous for the D187N mutation in gelsolin. In both, the disease was unusually severe, manifesting with nephrotic syndrome and end-stage renal failure. Immunohistochemical studies of the kidneys demonstrated heavy glomerular deposits of gelsolin-derived amyloid. Immunostaining also demonstrated gelsolin in the tubular epithelium, where it was Congo-red negative.

In 6 members of a family with Finnish-type amyloidosis, Potrc et al. (2021) identified a heterozygous missense mutation in the GSN gene (E580K; 137350.0003). The mutation, which was found by exome sequencing, segregated with disease in the family and was not present in the gnomAD database. The affected residue was located in the G5 domain, which is homologous to the second domain, which contains D187N (137350.0001), the most common pathogenic variant.

In 2 unrelated families segregating Finnish-type amyloidosis, one with a single affected person and the other with 3 first-degree relatives, Mullany et al. (2021) identified heterozygous mutations in the GSN gene. In the single affected person, the classic D187N mutation (also referred to as D214N) was identified. In the 3 affected family members (father, son, and daughter), a trp493-to-arg (W493R; 137350.0004) mutation was identified. The W493R variant was the first to be located in the G4 domain. Immunohistochemical studies on corneal tissue from the proband in the family with the W493R variant identified gelsolin protein within histologically defined corneal amyloid deposits.

See Also:

Meretoja (1973)

REFERENCES

1. Akiya, S., Nishio, Y., Ibi, K., Uozumi, H., Takahashi, H., Hamada, T., Onishi, A., Ishiguchi, H., Hoshii, Y., Nakazato, M. **Lattice corneal dystrophy type II associated with familial amyloid polyneuropathy type IV.** Ophthalmology 103: 1106-1110, 1996. [PubMed: 8684801] [Full Text: https://doi.org/10.1016/s0161-6420(96)30560-5]
2. Boysen, G., Galassi, G., Kamieniecka, Z., Schlaeger, J., Trojaborg, W. **Familial amyloidosis with cranial neuropathy and corneal lattice dystrophy.** J. Neurol. Neurosurg. Psychiat. 42: 1020-1030, 1979. [PubMed: 228009] [Full Text: https://doi.org/10.1136/jnnp.42.11.1020]
3. Darras, B. T., Adelman, L. S., Mora, J. S., Rodziner, R. A., Munsat, T. L. **Familial amyloidosis with cranial neuropathy and corneal lattice dystrophy.** Neurology 36: 432-435, 1986. [PubMed: 3513049] [Full Text: https://doi.org/10.1212/wnl.36.3.432]
4. de la Chapelle, A., Kere, J., Sack, G. H., Jr., Tolvanen, R., Maury, C. P. J. **Familial amyloidosis, Finnish type: G654-to-A mutation of the gelsolin gene in Finnish families and an unrelated American family.** Genomics 13: 898-901, 1992. [PubMed: 1322359] [Full Text: https://doi.org/10.1016/0888-7543(92)90182-r]
5. Haltia, M., Ghiso, J., Prelli, F., Gallo, G., Kiuru, S., Somer, H., Palo, J., Frangione, G. **Amyloid in familial amyloidosis, Finnish type, is antigenically and structurally related to gelsolin.** Am. J. Path. 136: 1223-1228, 1990. [PubMed: 2162627]
6. Haltia, M., Levy, E., Meretoja, J., Fernandez-Madrid, I., Koivunen, O., Frangione, B. **Gelsolin gene mutation--at codon 187--in familial amyloidosis, Finnish: DNA-diagnostic assay.** Am. J. Med. Genet. 42: 357-359, 1992. [PubMed: 1311149] [Full Text: https://doi.org/10.1002/ajmg.1320420321]
7. Kiuru, S., Matikainen, E., Kupari, M., Haltia, M., Palo, J. **Autonomic nervous system and cardiac involvement in familial amyloidosis, Finnish type (FAF).** J. Neurol. Sci. 126: 40-48, 1994. [PubMed: 7836945] [Full Text: https://doi.org/10.1016/0022-510x(94)90092-2]
8. Kiuru, S. **Familial amyloidosis of the Finnish type (FAF): a clinical study of 30 patients.** Acta Neurol. Scand. 86: 346-353, 1992. [PubMed: 1333716] [Full Text: https://doi.org/10.1111/j.1600-0404.1992.tb05099.x]
9. Klaus, E., Freyberger, E., Kavka, G., Vodicka, F. **Familial occurrence of a bulbar paralytic form of amyotrophic lateral sclerosis with reticular corneal dystrophy and cutis hyperelastica in 3 sisters.** Psychiat. Neurol. (Basel) 138: 79-97, 1959. [PubMed: 14409602]
10. Klintworth, G. K. **Lattice corneal dystrophy: an inherited variety of amyloidosis restricted to the cornea.** Am. J. Path. 50: 371-399, 1967. [PubMed: 4163628]
11. Maury, C. P. J., Alli, K., Baumann, M. **Finnish hereditary amyloidosis: amino acid sequence homology between the amyloid fibril protein and human plasma gelsoline.** FEBS Lett. 260: 85-87, 1990. [PubMed: 2153578] [Full Text: https://doi.org/10.1016/0014-5793(90)80072-q]
12. Maury, C. P. J., Kere, J., Tolvanen, R., de la Chapelle, A. **Finnish hereditary amyloidosis is caused by a single nucleotide substitution in the gelsolin gene.** FEBS Lett. 276: 75-77, 1990. [PubMed: 2176164] [Full Text: https://doi.org/10.1016/0014-5793(90)80510-p]
13. Maury, C. P. J. **Homozygous familial amyloidosis, Finnish type: demonstration of glomerular gelsolin-derived amyloid and non-amyloid tubular gelsolin.** Clin. Nephrol. 40: 53-56, 1993. [PubMed: 8395367]
14. Meretoja, J., Hollmen, T., Meretoja, T., Penttinen, R. **Partial characterization of amyloid proteins in inherited amyloidosis with lattice corneal dystrophy and in secondary amyloidosis.** Med. Biol. 56: 17-22, 1978. [PubMed: 305513]
15. Meretoja, J. **Genetic aspects of familial amyloidosis with corneal lattice dystrophy and cranial neuropathy.** Clin. Genet. 4: 173-185, 1973. [PubMed: 4543600] [Full Text: https://doi.org/10.1111/j.1399-0004.1973.tb01140.x]
16. Meretoja, J. **Inherited Systemic Amyloidosis with Lattice Corneal Dystrophy.** Acad. Dissertation: Helsinki (pub.) 1973.
17. Mullany, S., Souzeau, E., Klebe, S., Zhou, T., Knight, L. S. W., Qassim, A., Berry, E. C., Marshall, H., Hussey, M., Dubowsky, A., Breen, J., Hassall, M. M., Mills, R. A., Craig, J. E., Siggs, O. M. **A novel GSN variant outside the G2 calcium-binding domain associated with amyloidosis of the Finnish type.** Hum. Mutat. 42: 818-826, 2021. [PubMed: 33973672] [Full Text: https://doi.org/10.1002/humu.24214]
18. Nikoskinen, T., Schmidt, E.-K., Strbian, D., Kiuru-Enari, S., Atula, S. **Natural course of Finnish gelsolin amyloidosis.** Ann. Med. 47: 506-511, 2015. [PubMed: 26339870] [Full Text: https://doi.org/10.3109/07853890.2015.1075063]
19. Paunio, T., Kiuru, S., Hongell, V., Mustonen, E., Syvanen, A.-C., Bengtstrom, M., Palo, J., Peltonen, L. **Solid-phase minisequencing test reveals asp187-to-asn (G654-to-A) mutation of gelsolin in all affected individuals with Finnish type of familial amyloidosis.** Genomics 13: 237-239, 1992. [PubMed: 1315718] [Full Text: https://doi.org/10.1016/0888-7543(92)90235-k]
20. Potrc, M., Volk, M., de Rosa, M., Pizem, J., Teran, N., Jaklic, H., Maver, A., Drnovsek-Olup, B., Bollati, M., Vogelnik, K., Hocevar, A., Gornik, A., Pfeifer, V., Peterlin, B., Hawlina, M., Fakin, A. **Clinical and histopathological features of gelsolin amyloidosis associated with a novel GSN variant p.Glu580Lys.** Int. J. Molec. Sci. 22: 1084, 2021. [PubMed: 33499149] [Full Text: https://doi.org/10.3390/ijms22031084]
21. Purcell, J. J., Jr., Rodrigues, M., Chishti, M. I., Riner, R. N., Dooley, J. M. **Lattice corneal dystrophy associated with familial systemic amyloidosis (Meretoja's syndrome).** Ophthalmology 90: 1512-1517, 1983. [PubMed: 6610849] [Full Text: https://doi.org/10.1016/s0161-6420(83)34369-4]
22. Sack, G. H., Jr., Dumars, K. W., Gummerson, K. S., Law, A., McKusick, V. A. **Three forms of dominant amyloid neuropathy.** Johns Hopkins Med. J. 149: 239-247, 1981. [PubMed: 6975851]
23. Schmidt, E.-K., Mustonen, T., Kiuru-Enari, S., Kivela, T. T., Atula, S. **Finnish gelsolin amyloidosis causes significant disease burden but does not affect survival: FIN-GAR phase II study.** Orphanet J. Rare Dis. 15: 19, 2020. [PubMed: 31952544] [Full Text: https://doi.org/10.1186/s13023-020-1300-5]

24. Sipila, K., Aula, P. **Database for the mutations of the Finnish disease heritage.** Hum. Mutat. 19: 16-22, 2002. [PubMed: 11754099] [Full Text: https://doi.org/10.1002/humu.10019]

25. Starck, T., Kenyon, K. R., Hanninen, L. A., Beyer-Machule, C., Fabian, R., Gorn, R. A., McMullan, F. D., Baum, J., McAdam, K. P. W. J. **Clinical and histopathologic studies of two families with lattice corneal dystrophy and familial systemic amyloidosis (Meretoja syndrome).** Ophthalmology 98: 1197-1206, 1991. [PubMed: 1923356] [Full Text: https://doi.org/10.1016/s0161-6420(91)32153-5]

26. Winkelman, J. E., Delleman, J. W., Ansink, B. J. J. **Ein hereditaeres Syndrom, bestehend aus peripherer Polyneuropathie, Hautveraenderungen und gittriger Dystrophie der Hornhaut.** Klin. Monatsbl. Augenheilkd. 159: 618-623, 1971. [PubMed: 4109360]

Contributors: Anne M. Stumpf - updated : 05/20/2024
Sonja A. Rasmussen - updated : 08/22/2023

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